Practice Set 3 Paper 3 – Subspecialities of Urology

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Practice paper compiled from the question bank.

Q1. Describe in brief urological complications of OBGY surgery. Describe in detail, management of urethrovaginal fistula

Type: LAQ (25 marks) (urethrovaginal fistula also answered in 2025-Winter Paper 3, Q3a; iatrogenic ureteric injury in Practice Set 2 Paper 3, Q1)

Assumed meaning: OBGY = obstetric and gynaecological surgery.

Part A: Urological complications of OBGY surgery

- The bladder, distal ureter and urethra lie against the cervix, uterine vessels and anterior vaginal wall. Risk rises with distorted anatomy: fibroids, endometriosis, cancer, previous caesarean, radiation, heavy bleeding, prolapse or mesh surgery.

Complication	Key facts
Bladder injury (commonest organ injured)	Caesarean 0.27–0.3%; hysterectomy about 0.3–0.75% (laparoscopic highest); retropubic tape 3–9%. Repair: debride, 2-layer closure, fill test, catheter 7–14 days; check ureters if trigonal
Ureteric injury	Gynaecology causes over half of iatrogenic injuries; over 90% distal third. Hysterectomy: vaginal 0.02–0.5%, abdominal 0.03–2%, minimally invasive 0.13–6%; prolapse surgery 1.7–4.3%. **Ligation** is the commonest gynaecological mechanism
Fistulae	VVF (vesicovaginal fistula), about 0.8 per 1000 hysterectomies; UVF (ureterovaginal fistula); vesicouterine fistula after caesarean (Youssef syndrome); urethrovaginal fistula
Voiding dysfunction	Radical hysterectomy (pelvic plexus injury) 16–80% historically: retention, poor compliance; CIC (clean intermittent catheterisation) and wait. Post-partum or post-epidural overdistension
Mesh and sling	Exposure, erosion into bladder or urethra, outlet obstruction, pain, dyspareunia
Radiation	Radiation cystitis; VVF in 0.6–2% after cervical cancer radiotherapy; ureteric stricture

- **Ureteric danger sites:** pelvic brim (infundibulopelvic ligament), under the uterine artery, beside the uterosacral ligament, anterolateral vaginal fornix.
- **Prevention and recognition:** haemostasis and anatomy; prophylactic stents only selectively (they aid recognition but do not prevent injury); cystoscopy with dye (IV indigo carmine or oral phenazopyridine) after minimally invasive hysterectomy and prolapse or continence surgery.
- Only about one-third of ureteric injuries are recognised at open surgery. Missed injury presents with fever, ileus, flank pain, urinoma or leakage; do CTU (computed tomography urography) with delayed films, then retrograde stenting. Distal repair: ureteroneocystostomy ± psoas hitch or Boari flap (see Q3d).

Part B: Urethrovaginal fistula

Aetiology

- Developing world: **obstructed labour**, often with a VVF.
- Developed world: **urethral diverticulectomy** (fistula in 1–8%; causes 20–25% of urethrovaginal fistulae), slings, bulking agents, anterior repair or mesh, indwelling catheter pressure necrosis, pessaries, radiation, trauma.

Clinical features

- Depend on position relative to the sphincter. **Distal** fistula: often asymptomatic, split stream, vaginal voiding, post-void dribble. **Proximal** (bladder neck): continuous leakage or

SUI (stress urinary incontinence).
- Dyspareunia, recurrent UTI (urinary tract infection), perineal irritation; partner discomfort with exposed mesh.

Evaluation

- **Speculum examination** usually suffices; methylene blue test if rugae hide it (distal stain on tampon).
- **VCUG (voiding cystourethrogram), lateral view;** VUDS (videourodynamics) for coexisting SUI or detrusor overactivity.
- **Cystourethroscopy** (flexible is better, outflow at the lens): size, site, bladder neck, mesh. **About 20% have a separate VVF.** Image the upper tract.
- Goh location: type 1 >3.5 cm, 2: 2.5–3.5, 3: 1.5–2.5, 4: <1.5 cm from meatus. Closer means worse continence.
- Optimise: treat UTI, vaginal oestrogen, nutrition; biopsy if previous cancer.

Management

- Asymptomatic distal fistula: may need no repair. Timing: early if tissue healthy; otherwise after inflammation settles.
- **Standard: transvaginal rotational-flap repair:**

Step	Details
1. Exposure	Lithotomy (prone gives better anterior view); cystourethroscopy, cannulate tract; suprapubic catheter (Lowsley tractor), urethral catheter
2. Incision	Hydrodissect; circumscribe tract with 1–2 mm cuff, minimal excision; inverted **J or U** incision
3. Flaps	Raise vaginal flap; mobilise periurethral tissue to bladder neck; open retropubic space if sling planned
4. Closure	Over a urethral sound: **4-0 absorbable**, transverse or longitudinal; **second layer perpendicular**; buccal graft or flap onlay for large defects; leak test
5. Interposition	**Martius flap** (labial fat pad, external pudendal pedicle) most often; omentum, gracilis, rectus fascia occasionally
6. Cover	Advance vaginal flap beyond closure (2-0)

- **Autologous pubovaginal sling** for SUI or bladder-neck involvement; **never synthetic** (erodes ischaemic urethra). Remove all mesh in the field.
- **Near-total urethral loss:** tubularised anterior bladder flap neourethra. Irreparable cases: diversion.
- **After repair:** urethral catheter out at 2–3 days; suprapubic ≥14 days; lateral VCUG before removal; nothing per vagina for 6 weeks.

Outcomes and complications

- Closure **90–95%**, but continence only **50–80%**; postoperative SUI up to half. Sling then or staged later.
- Urethral stricture, recurrence, dyspareunia.

Diagram to draw: inverted-U vaginal flap, two perpendicular urethral layers, Martius flap.

> **Key points to remember:**
> - Bladder is the commonest organ injured; ureteric injury is mostly distal and missed.
> - Diverticulectomy is the commonest Western cause; 20% have a coexisting VVF.
> - Layered closure, Martius flap, autologous sling only.
> - Closure 90–95%; continence only 50–80%.

Q2. Discuss management options of VVF

Type: LAQ (25 marks) (timing in 2021-Summer Paper 2, Q3b; recurrent VVF in 2025-Winter Paper 2, Q3f)

Introduction

- VVF (vesicovaginal fistula): epithelialised bladder–vagina tract; commonest acquired urinary fistula (>75% of genitourinary fistulae).
- **Developing world:** obstetric (>90%); fetal head impacted for days causes pressure necrosis. **Developed world:** iatrogenic (>75% after hysterectomy), usually supratrigonal at the vaginal cuff. Also radiation, malignancy, mesh, foreign body.
- Mechanism: ischaemic necrosis; continuous urine flow keeps the tract open until it epithelialises.

Classification

Goh (distance from meatus)	Size	Tissue
1: >3.5 cm; 2: 2.5–3.5; 3: 1.5–2.5; 4: <1.5 cm	A <1.5 cm; B 1.5–3 cm; C >3 cm	i mild fibrosis; ii moderate/severe; iii radiation, ureteric, circumferential, previous repair

- **Also know:** Waaldijk (closing mechanism involvement); WHO (World Health Organization) simple vs complex.

Evaluation (guides choice)

- Speculum; fluid creatinine. **Dye tests:** single (blue backfill, tampon); **double** (oral phenazopyridine plus blue): orange = UVF (ureterovaginal fistula), blue = VVF.
- **Cystoscopy and vaginoscopy:** number, size, ureteric orifices, bladder neck, mesh; **biopsy if previous cancer.**
- **Upper tract (CTU, computed tomography urography) is mandatory:** coexisting ureteric fistula (up to 12–25%).
- Urodynamics: capacity, compliance, SUI (stress urinary incontinence) in 10–15%.
- Prepare: vaginal oestrogen for atrophy; counsel about residual incontinence, dyspareunia and possible reoperation.

Principles

- Optimise nutrition, treat infection, drain continuously, relieve obstruction, exclude malignancy.
- Surgery: tension-free, multilayer, **non-overlapping**, watertight closure; remove foreign material; consider interposition; drain afterwards. **The first repair has the best chance.**

Option 1: Conservative and endoscopic

- **Catheter drainage** cures about 15%. Favourable: <1 cm (best <3 mm), started within **3 weeks**, non-irradiated. Unlikely after 6 weeks (epithelialised). EAU (European Association of Urology): if tried, ≥12 weeks.
- **Fulguration** plus 2 weeks' drainage for tracts <3.5 mm.
- **Sealants** (fibrin glue, cyanoacrylate, collagen): tiny series; not recommended by EAU.

Timing

- Clean iatrogenic: early repair (EAU: no minimum delay if no infection or recent radiation). Obstetric with necrosis: about 3 months. Radiation: **≥12 months.** Some avoid 4–12 weeks.

Option 2: Surgical route

Vaginal	Abdominal (open, laparoscopic, robotic)
Low, trigonal, accessible apex	Supratrigonal, narrow vagina, no descent
Ureters distant	Ureter involved, reimplant needed
Less pain, blood loss, stay	Augmentation, Tanagho flap, diversion possible
Risk: vaginal shortening	Large (≥4 cm primary, ≥2 cm recurrent), radiation

- First-attempt closure about **90% either route**; choose the route the surgeon knows best.

Vaginal techniques

- **Flap-splitting (Raz):** ureteric stents if within 1 cm; Foley through tract for traction; inverted J/U incision; raise flaps; layer 1 3-0/4-0 including tract edge; layer 2 perpendicular; fill test 200–300 mL; Martius or peritoneal flap; advance opposite flap ≥2 cm beyond.
- **Latzko (high partial colpocleisis):** denude vagina 1–2 cm around tract and close vaginal walls over it; bladder not opened. ≥90% in non-irradiated VVF; favoured for radiation VVF and tracts near the ureters; vaginal shortening.
- **Also know:** vaginal cuff excision technique.

Abdominal techniques

- **O'Conor (1951):** extraperitoneal; bivalve bladder from dome down to fistula; excise tract; separate bladder from vagina 2–3 cm beyond; close vagina; **omental or peritoneal flap**; close bladder; suprapubic catheter.
- **Transvesical** (limited cystotomy) and **extravesical** (no cystotomy; fewer LUTS [lower urinary tract symptoms]).
- **Laparoscopic (1994) and robotic (2005):** same closure (85–100%) with less blood loss and shorter stay; vaginal stent, backfill, ICG (indocyanine green) perfusion check.

Interposition flaps (for radiation, recurrent, ischaemic or large VVF)

Flap	Supply	Use
Martius (labial fat pad)	Internal and external pudendal	Mid–distal VVF; not apex
Peritoneal (vaginal)	Pouch of Douglas fold	Apical post-hysterectomy
Omentum (J-flap)	Gastroepiploic arteries	Abdominal; radiation, large
Gracilis, Singapore, rectus	Myocutaneous	Refractory, tissue loss

- RCT (randomised controlled trial): Martius added nothing in simple VVF, with donor-site pain in nearly a third.

Postoperative care

- Catheter ≥10–14 days (urethral and/or suprapubic); antimuscarinic or β3-agonist for spasm; cystogram before removal; stents out at 4–6 weeks.

Option 3: Urinary diversion

- Multiple failures, severe radiation, massive tissue loss: conduit or continent reservoir. Unfit: ureteric occlusion with nephrostomies. Resource-poor: ureterosigmoidostomy as last resort.

Outcomes

- Residual urgency or incontinence despite closure (Goh 3–4 predicts); retention (vertical bladder closure); dyspareunia (narrow vagina, fistula ≥3 cm). ICS (International Continence Society): pelvic floor physiotherapy.

Diagram to draw: flap-splitting repair: perpendicular bladder layers, Martius flap, advanced vaginal flap.

Key points to remember:

- Image upper tracts; exclude ureteric fistula and cancer.
- Catheter trial only if small, fresh (<3 weeks), non-irradiated.
- Vaginal and abdominal routes both close about 90%.
- Interposition for complex VVF; radiation repair after ≥12 months.

Q3a. Discuss diagnostics and operative pediatric laparoscopy in urology

Type: SAQ (10 marks)

Evolution

- Diagnostic (non-palpable testis, 1976) → nephrectomy (1994) → pyeloplasty (1995) → robotic (2002).

Diagnostic laparoscopy

- **Non-palpable testis** (procedure of choice, after examination under anaesthesia): note vessels, vas, testis and internal ring.

Finding	Meaning
Blind-ending vessels and vas, closed ring	Vanishing testis: stop
Vessels and vas entering ring	Explore groin; excise remnant
Intra-abdominal testis	Orchidopexy (primary or Fowler-Stephens)

- **DSD (differences of sex development):** gonadal inspection and biopsy, Müllerian structures.

Operative laparoscopy

- **Testis:** primary orchidopexy (>90% success); Fowler-Stephens one- or two-stage; gonadectomy in DSD.
- **Kidney:** nephrectomy, heminephrectomy, **pyeloplasty** (commonest MIS (minimally invasive surgery) procedure).
- **Ureter and bladder:** extravesical or vesicoscopic (cross-trigonal) reimplantation, Mitrofanoff, augmentation, bladder neck reconstruction, utricle and urogenital sinus.
- **Also know:** RPLND (retroperitoneal lymph node dissection) for paratesticular rhabdomyosarcoma.

Technical principles

- **Access:** open Hasson or Veress at umbilicus (Palmer's point if scarred); equal safety.
- 3–5 mm ports; robots use 8 mm. Transperitoneal (most space), retroperitoneal (spares peritoneum, useful in peritoneal dialysis), vesicoscopic.
- **Pressures:** CO_2 absorption, reduced venous return, raised airway pressure; reversible oliguria (no fluid bolus).

Age	0–1 month	2–12 months	1–2 years	2–8 years
Max IAP (intra-abdominal pressure)	≤6 mmHg	≤8	≤10	≤12

- **Positioning:** modified flank for kidney; supine steep Trendelenburg for pelvis. Infants: minimise inhalational anaesthetic, rectal tube to reduce bowel distension.
- **Only absolute contraindication:** inability to tolerate pneumoperitoneum.

Complications

- Right common iliac vessel injury (behind umbilicus; commonest major), inferior epigastric (commonest minor), bowel injury, subcutaneous emphysema, **port-site hernia 0.5–3%** (close fascia).

Pros and cons

- Shorter stay, better cosmesis, deep pelvic access; pain benefit unproven. Robot: cost, learning curve; infant working space only about 1 L.

Diagram to draw: laparoscopic view of internal ring with blind-ending vessels.

> **Key points to remember:**
>
> - Diagnostic laparoscopy is the procedure of choice for non-palpable testis.
> - Pressure ceiling 6–12 mmHg by age.
> - Right common iliac at Veress entry; close port fascia.

Q3b. Tadalafil

Type: SAQ (10 marks)

Pharmacology

- Selective **PDE5 (phosphodiesterase type 5) inhibitor**; FDA (Food and Drug Administration) approved 2003.
- **Mechanism:** sexual stimulation releases NO (nitric oxide), raising cGMP (cyclic guanosine monophosphate), which relaxes cavernosal smooth muscle. PDE5 breaks down cGMP; tadalafil blocks this. It **augments, not initiates,** erection.
- Structurally distinct from sildenafil; cross-reacts with **PDE11**, little with PDE6 (retina).

Property	Tadalafil (vs sildenafil)
Time to peak	2 h (0.8 h)
Half-life	**17.5 h** (3–5 h)
Fatty food	No effect (reduces absorption)
Window	Widest (up to 36 h)

Uses and doses

1. **ED (erectile dysfunction):** 10–20 mg on demand, or 2.5–5 mg once daily. Intercourse success about 70%; 40–50% in diabetes and after prostatectomy.
2. **BPH (benign prostatic hyperplasia) with LUTS (lower urinary tract symptoms), with or without ED: 5 mg daily**, the only PDE5 inhibitor licensed. IPSS (International Prostate Symptom Score) falls about 6 points (placebo 3.6) **without Qmax (peak flow rate) change**; mechanism is NO–cGMP relaxation, perfusion and reduced afferent activity, not de-obstruction. Equal to tamsulosin for IPSS. EAU (European Association of Urology) 2022 and AUA (American Urological Association) 2021: option for moderate–severe LUTS.
3. **Also know:** fixed finasteride–tadalafil (FDA 2021); penile rehabilitation (benefit unproven); prevention of stuttering priapism (PDE5I paradox); pulmonary arterial hypertension.

Adverse effects

- Headache, dyspepsia, **back pain and myalgia** (characteristic), flushing, nasal congestion. Visual effects rare.
- NAION (non-arteritic anterior ischaemic optic neuropathy): no proven link; stop if sudden visual loss.

Contraindications and interactions

- **Nitrates in any form: absolute.** After tadalafil, no nitrates for **48 h** (24 h for sildenafil).
- Caution: α-blockers (start lowest dose of the added drug), aortic stenosis, hypotension. CYP3A4 inhibitors (ketoconazole, ritonavir) raise levels.
- Not studied: MI (myocardial infarction) or stroke within 6 months, unstable angina, BP <90/50 or >170/100 mmHg, retinitis pigmentosa.

Diagram to draw: NO–cGMP–PDE5 pathway in cavernosal smooth muscle.

> **Key points to remember:**
> - Half-life 17.5 h; food-independent; back pain.
> - 5 mg daily for LUTS: IPSS improves, Qmax does not.
> - No nitrates for 48 h after a dose.

Q3c. DDAVP

Type: SAQ (10 marks) (also answered in 2023-Winter Paper 3, Q3b)

Pharmacology

- DDAVP (**1-deamino-8-D-arginine vasopressin**, desmopressin): synthetic analogue of AVP (arginine vasopressin).
- Deamination and D-arginine give longer action and **selective V2 agonism without pressor (V1) effect**.
- **Mechanism:** V2 receptor on collecting-duct cells raises cAMP (cyclic adenosine monophosphate), inserting **aquaporin-2**; water is reabsorbed, so urine is less and concentrated.
- Endothelial V2 action releases von Willebrand factor and factor VIII.

Formulations

Form	Dose
Tablet	0.2–0.4 mg, about 1 h before bed (enuresis)
Oral lyophilisate (melt)	120–240 µg
Low-dose nasal spray (nocturia)	0.83 or 1.66 µg, 30 min before bed
Sublingual ODT (orally disintegrating tablet)	27.7 µg women, 55.3 µg men
IV or subcutaneous	0.3 µg/kg (bleeding)

Uses

1. **Monosymptomatic nocturnal enuresis:** first line with the alarm. **Rule of thirds** (full, partial, no response). Works at once but relapses on stopping; avoid evening fluid; add oxybutynin if partial.
2. **Nocturia from nocturnal polyuria** (NPi [nocturnal polyuria index] >20–33%, age-dependent). EAU (European Association of Urology): offer under 65; low dose over 65 with ≥2 voids. Add to α-blocker for refractory nocturia in BPH (benign prostatic hyperplasia).
3. **Neurogenic nocturia:** multiple sclerosis.
4. **Bleeding:** uraemic platelet dysfunction, von Willebrand disease (including acquired, in Wilms tumour). Tachyphylaxis after repeated doses.
5. **Diagnosis:** central versus nephrogenic DI (diabetes insipidus).

Adverse effects

- **Hyponatraemia** (most important): headache, nausea, confusion, seizures.
- Risk factors: **age ≥65**, sodium <135 mmol/L, CrCl (creatinine clearance) <60 mL/min, higher dose; women more sensitive. Most events in the first **2–3 weeks**.

Monitoring and contraindications

- Sodium at baseline, within 1 week and at 1 month (EAU adds day 3), and after dose change.
- **Avoid:** hyponatraemia, polydipsia, heart failure, SIADH (syndrome of inappropriate antidiuretic hormone), significant renal impairment, **loop diuretics**.

Diagram to draw: V2 receptor–cAMP–aquaporin-2 pathway in the collecting duct.

Key points to remember:

- Selective V2 agonist; no pressor effect.
- Enuresis: 0.2–0.4 mg tablet or 120–240 µg melt; rule of thirds.
- Hyponatraemia: age ≥65, low sodium; check sodium early.

Q3d. Boari flap

Type: SAQ (10 marks)

Definition

- A **tubularised flap of anterior bladder wall**, based posterolaterally on the side of the lesion, swung up to bridge a distal or mid-ureteric gap. Described in dogs 1894, humans 1947; laparoscopic 2001, robotic 2008.

Reconstructive ladder (distal ureter)

Option	Gap bridged
Ureteroneocystostomy	Distal 4–5 cm
+ Psoas hitch	Adds up to 5 cm (total 6–10 cm)
Boari flap (+ hitch)	**10–15 cm**; spiral flap may reach pelvis (right)
Downward nephropexy	Adds up to 5–8 cm

- Beyond: TUU (transureteroureterostomy), ileal ureter, autotransplantation.

Indications

- Long distal or mid-ureteric loss: gynaecological injury (commonest), stricture, failed reimplantation, distal TCC (transitional cell carcinoma) ureter, tuberculosis.

Contraindications

- **Small, contracted or poorly compliant bladder** (radiation, tuberculosis); uncorrected outlet obstruction or neurogenic bladder. Not usually for acute unstable trauma.
- Preoperative: urodynamics or cystogram for capacity; antegrade and retrograde studies for length.

Technique

1. Mobilise bladder; divide contralateral superior vesical pedicle if needed.
2. Mark flap on the **non-distended** bladder: **base ≥4 cm, apex ≥3 cm, length:base ≤3:1**; length = gap + 3 cm if tunnelled.
3. Raise flap; **psoas hitch** its base (sutures superficial, parallel to fibres: spare genitofemoral and femoral nerves).
4. Anastomose spatulated ureter: **non-refluxing** 3 cm submucosal tunnel, or **refluxing** direct (apex about 3 times ureteric diameter).
5. Stent (double-J); tubularise in two layers 2-0 absorbable; drain, catheter 7–14 days.

- Laparoscopic and robotic: transperitoneal, refluxing anastomosis, results excellent.

Complications

- **Recurrent stricture** (commonest; ischaemia or tension), reduced capacity with frequency and urgency, leak, nerve palsy, iliac vessel injury, pseudodiverticulum.
- eGFR (estimated glomerular filtration rate) fell in 44% within 2 years after reimplantation: counsel.
- **Alternative:** bladder advancement flap, similar reach and simpler.

Diagram to draw: bladder outline with oblique flap (4 cm base, 3 cm apex), hitched and tubularised to ureter.

Key points to remember:

- Bridges 10–15 cm; combine with psoas hitch.
- Base ≥4 cm, length:base ≤3:1, measure bladder empty.
- Contraindicated in small contracted bladder.

Q3e. Single stage repair of Extrophy of UB.

Type: SAQ (10 marks)

Assumed meaning: single-stage (complete primary) repair of exstrophy of the urinary bladder, mainly CPRE (complete primary repair of exstrophy, Mitchell).

Background

- Classic bladder exstrophy: absent infraumbilical mesoderm; open bladder, epispadias, pubic diastasis (mean 4.8 cm).
- **MSRE (modern staged repair):** closure at birth, epispadias repair 6–12 months, bladder neck reconstruction 4–7 years. Single-stage repairs aim for fewer operations and continence without bladder neck repair.

Single-stage options

Repair	Features
CPRE (Mitchell)	Newborn closure + penile disassembly + bladder neck tailoring
Erlangen ("total")	All in one at about 8 weeks, no osteotomy
Combined closure + epispadias repair	Delayed or failed closures, boys >6 months

Selection

- Template ≥3 × 3 cm, elastic, few polyps; experienced team. Not for penoscrotal duplication, hypoplastic bladder, bilateral hydronephrosis.

CPRE technique

1. Combined general and epidural anaesthesia; **osteotomy if >72 hours** (3 days) old or diastasis >4 cm or stiff pelvis.
2. Incise around plate; free bladder; **divide urogenital diaphragm fibres** to levator hiatus so the unit sits deep.
3. **Penile disassembly:** right and left hemicorporoglanular bodies plus urethral wedge on spongiosum (now often left attached to glans).
4. Urethra placed ventrally between corpora; corpora joined; meatus usually **hypospadiac**.
5. Bladder and neck closed and tapered; ± bilateral ureteric reimplantation (BUR-CPRE).
6. Pubic No. 2 nylon mattress suture; suprapubic catheter 4 weeks, ureteric stents.
7. Immobilise: external fixation plus traction (failure 7.5% vs spica 47%).

Complications

- Same frequency as MSRE but **more serious**: dehiscence, prolapse, vesicocutaneous fistula, **glans or corporal loss** (ischaemia), early reimplantation for VUR (vesicoureteric reflux) or hydronephrosis; retention in girls.

Outcomes

- Most still need bladder neck surgery. Daytime dryness: about 56% after CPRE with YDL (Young-Dees-Leadbetter), versus 70–74% after MSRE.
- Continence predictors: **template size and successful first closure**.

Diagram to draw: penile disassembly into two hemicorpora and urethral wedge.

Key points to remember:

- CPRE: closure + penile disassembly in newborn.
- Osteotomy if >72 h or diastasis >4 cm.
- Risks: glans loss, hypospadias, VUR.
- First closure success predicts continence.

Q3f. Retrograde ejaculation

Type: SAQ (10 marks)

Definition

- RE (retrograde ejaculation): semen passes backwards into the bladder at orgasm because the **bladder neck fails to close**; orgasm is preserved, ejaculate absent or low volume.

Physiology

Phase	Event	Control
Emission	Vas, seminal vesicles, prostate deliver semen to posterior urethra	Sympathetic T10–L2
Ejection	**Bladder neck closes**, external sphincter relaxes, bulbospongiosus contracts	Somatic S2–S4 + sympathetic reflex
Orgasm	Cerebral perception	Cortical

Causes

- **Anatomical: TURP (transurethral resection of prostate)** 42–100%, TUIP (incision) 5–45%; most other resections and enucleations; previous bladder neck reconstruction. Ejaculation-sparing techniques, PUL (prostatic urethral lift) and Aquablation preserve ejaculation.
- **Neurogenic: diabetic autonomic neuropathy** (commonest medical), RPLND (retroperitoneal lymph node dissection; less with nerve sparing), colorectal surgery, sympathectomy, aortic surgery, spinal cord injury, multiple sclerosis.
- **Drugs:** α-blockers (silodosin, tamsulosin), antipsychotics, antidepressants, thiazides.
- **Congenital:** Müllerian duct cyst, Wolffian duct anomaly, prune belly syndrome.

Evaluation

- Low-volume or absent ejaculate with orgasm; infertility.
- **Post-ejaculatory urinalysis:** void, ejaculate, void again; centrifuge. **>5–10 sperm per HPF (high-power field)** confirms RE; significant if urinary sperm approach or exceed antegrade sperm.
- Sperm and fructose in urine separate RE from **failure of emission**; TRUS (transrectal ultrasound) if EDO (ejaculatory duct obstruction) suspected.

Treatment

- Reassure if fertility not desired; stop offending drug.
- **Sympathomimetics** (close bladder neck): **pseudoephedrine** 60–120 mg (120 mg 6-hourly the day before and twice on the day); midodrine, ephedrine, phenylpropanolamine. Side effects: dry mouth, hypertension.
- **Imipramine** 25 mg twice daily; antihistamines (brompheniramine) add to sympathomimetics.
- **Bladder neck reconstruction:** poor results; collagen injection experimental.
- **Sperm retrieval:** alkalinise urine (sodium bicarbonate), void or catheterise after orgasm, wash sperm in buffer → IUI (intrauterine insemination), IVF (in vitro fertilisation) or ICSI (intracytoplasmic sperm injection).
- **Prevention:** nerve-sparing RPLND, ejaculation-preserving procedures for BPH (benign prostatic hyperplasia).

Diagram to draw: bladder neck open at ejaculation with semen entering the bladder.

Key points to remember:

- TURP and diabetic neuropathy are the commonest causes.
- Post-ejaculatory urine: sperm and fructose confirm RE.
- Pseudoephedrine first; else urinary sperm retrieval for assisted reproduction.

Rapid Revision Sheet – Practice Set 3 Paper 3

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Describe in brief urological complications of OBGY surgery. Describe in detail, management of urethrovaginal fistula (LAQ)

- Bladder is the commonest organ injured; ureteric injury is mostly distal and missed.
- Diverticulectomy is the commonest Western cause; 20% have a coexisting VVF.
- Layered closure, Martius flap, autologous sling only.
- Closure 90–95%; continence only 50–80%.

Q2. Discuss management options of VVF (LAQ)

- Image upper tracts; exclude ureteric fistula and cancer.
- Catheter trial only if small, fresh (<3 weeks), non-irradiated.
- Vaginal and abdominal routes both close about 90%.
- Interposition for complex VVF; radiation repair after ≥12 months.

Q3a. Discuss diagnostics and operative pediatric laparoscopy in urology (SAQ)

- Diagnostic laparoscopy is the procedure of choice for non-palpable testis.
- Pressure ceiling 6–12 mmHg by age.
- Right common iliac at Veress entry; close port fascia.

Q3b. Tadalafil (SAQ)

- Half-life 17.5 h; food-independent; back pain.
- 5 mg daily for LUTS: IPSS improves, Qmax does not.
- No nitrates for 48 h after a dose.

Q3c. DDAVP (SAQ)

- Selective V2 agonist; no pressor effect.
- Enuresis: 0.2–0.4 mg tablet or 120–240 µg melt; rule of thirds.
- Hyponatraemia: age ≥65, low sodium; check sodium early.

Q3d. Boari flap (SAQ)

- Bridges 10–15 cm; combine with psoas hitch.
- Base ≥4 cm, length:base ≤3:1, measure bladder empty.
- Contraindicated in small contracted bladder.

Q3e. Single stage repair of Extrophy of UB. (SAQ)

- CPRE: closure + penile disassembly in newborn.
- Osteotomy if >72 h or diastasis >4 cm.
- Risks: glans loss, hypospadias, VUR.
- First closure success predicts continence.

Q3f. Retrograde ejaculation (SAQ)

- TURP and diabetic neuropathy are the commonest causes.
- Post-ejaculatory urine: sperm and fructose confirm RE.
- Pseudoephedrine first; else urinary sperm retrieval for assisted reproduction.